

Wireless Networking

Task 1 :- List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

ANS :-

Wi-Fi 4 (802.11n)

- **Key Difference :** Introduced dual-band support (2.4 GHz & 5 GHz) with speeds up to 600 Mbps.

Wi-Fi 5 (802.11ac)

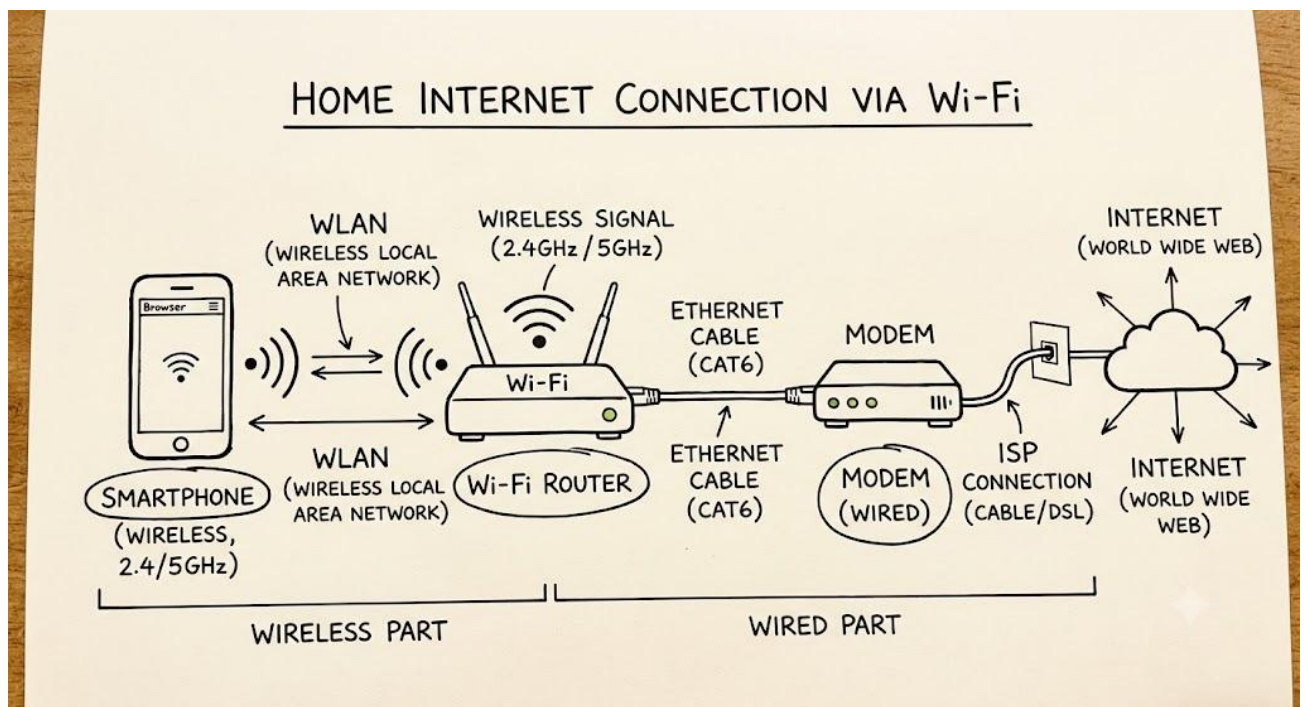
- **Key Difference :** Runs almost exclusively on the faster 5 GHz band, boosting speeds up to ~3.5 Gbps.

Wi-Fi 6 (802.11ax)

- **Key Difference :** Uses 2.4 GHz and 5 GHz much more efficiently, reaching top speeds up to 9.6 Gbps while handling busy networks better.

Task 2 :- Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.

ANS :-



Task 3 :- Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.

ANS :-

Scanned Networks :-

- **Network 1 (Home_WiFi_5G)**
 - Signal: Strong (-45 dBm / 4 bars)
 - Security: WPA3

- **Network 2 (Neighbor_Guest)**
 - Signal: Medium (-68 dBm / 2 bars)
 - Security: WPA2

- **Network 3 (Free_Public_WiFi)**
 - Signal: Strong (-50 dBm / 4 bars)
 - Security: Open (No password)

Which one to choose?

Choose: Network 1 (Home_WiFi_5G)

Why :-

- **Max Security:** Uses WPA3, which is the newest and safest protection against hackers.
- **Best Speed:** Has the strongest signal, meaning fast and stable internet without lag.

Task 4 :- Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

ANS :-

SSID hiding means turning off your router's name broadcast so your Wi-Fi network doesn't show up in the list of nearby networks. To connect, people must manually type in the exact name.

- **Advantage :**

Fewer random connection attempts. Strangers passing by won't see your network, so they won't try to guess your password or ask to use it.

- **Disadvantage :**

False sense of security (and connection hassle). Free hacker tools can still easily spot hidden networks, and you have to manually enter the network name every time you set up a new device.